

Name

Roll No.:

General Instructions:

1. Read the questions carefully.
2. Attempt all the questions.
3. No negative marking.
4. Answers to be marked in the OMR sheet. Darken the bubble completely. Partially darkened bubbles will not be considered.
5. Question number 1 to 100 carries 1 mark each

Read the passage and answer the questions (1-3) which follow:

The question of why women tend to live longer than men has intrigued scientists, psychologists, and social thinkers for generations. Although the reasons are complex, most researchers agree that a combination of biological, behavioural, and social factors contributes to this persistent difference in life expectancy.

From a biological standpoint, women appear to possess certain natural advantages. Their bodies generally show greater resistance to disease and recover more efficiently from illness. Genetic differences play a significant role: women have two X chromosomes, while men have one X and one Y. The presence of a second X chromosome provides a kind of genetic safety net, protecting against potentially harmful mutations. Hormonal factors further add to this advantage, as oestrogen is known to support cardiovascular health and strengthen the immune system.

However, biology explains only part of the story. Behaviour and lifestyle choices strongly influence longevity. Men are more likely to engage in high-risk behaviours such as smoking, drinking, or reckless driving. Such habits, often encouraged by cultural expectations of toughness or independence, may bring short-term thrill but long-term harm. Additionally, men are less likely to visit doctors regularly or seek preventive healthcare, which means that illnesses often go undetected until they become serious.

Social patterns and emotional habits also play a significant part. Traditionally, women have taken on greater caregiving roles, caring for children, parents, or communities. These responsibilities, while demanding, often foster empathy, social connection, and a sense of purpose — all linked to better mental and physical health. Women also tend to maintain stronger social networks and express their emotions more openly, reducing stress and promoting psychological well-being. These qualities may collectively contribute to their longer lives.

Encouragingly, the longevity gap is not fixed. As gender roles evolve, more men are adopting healthier routines, participating in caregiving, and prioritising self-care. Experts suggest that men who “borrow” habits often associated with women — such as balanced eating, regular medical check-ups, and open emotional communication — can significantly improve both their quality of life and lifespan. Governments and health organisations are also promoting preventive care and mental health awareness among men to bridge this gap further.

Ultimately, the question of longevity is not about competition but understanding. It reminds us that the length and quality of life depend on more than genetics; they are deeply influenced by habits, attitudes, and social behaviour. By learning from one another and adopting balanced lifestyles, both men and women can work toward a healthier, longer, and more fulfilling life. Longevity, then, becomes not just a biological gift, but a conscious choice shaped by the way we live.

1. Which of the following statements is NOT supported by the passage?

- A) Women's longer lifespan is influenced by both biological and social factors.
- B) Men's reluctance to seek preventive healthcare contributes to shorter lifespans.
- C) The longevity gap between men and women is an unchangeable biological reality.
- D) Evolving gender roles have encouraged men to adopt healthier lifestyle habits.

2. Read the following two statements and answer the question that follows:

Statement I: The study suggests that the female advantage in longevity dates back millions of years and is rooted in evolutionary patterns across species.

Statement II: Men can completely eliminate the longevity gap by adopting healthier lifestyles and behaviors.

Which of the following best describes the relationship between the two statements?

- A) Statement II oversimplifies the implications of Statement I, which emphasize evolutionary influences.
- B) Statement I provides partial support for the optimism expressed in Statement II.
- C) Statement II contradicts Statement I by denying the role of evolution in longevity differences.
- D) Statement I and Statement II present independent explanations that are both equally valid.

3. Which of the following best describes the tone and purpose of the passage?

- A) Analytical and persuasive — it presents evidence and encourages behavioural change.
- B) Descriptive and emotional — it praises women's superiority over men.
- C) Narrative and reflective — it recounts personal experiences about longevity.
- D) Critical and argumentative — it challenges scientific explanations of gendered lifespan.

Read the passage and answer the questions(Q4-6) which follow: (301 words)

In February 2022, a cyberattack on a leading Japanese automobile manufacturer forced the shutdown of five assembly plants for nearly two days. Investigators later discovered that the breach began with a malware-infected supplier system, halting production of more than 13 000 vehicles and causing losses of about \$250 million. What seemed like a small technical glitch exposed a critical truth: in the digital economy, a single compromised node can disrupt an entire network.

This incident reflects the paradox of modern efficiency. The "just-in-time" supply model, designed to reduce storage costs and speed up delivery, depends on seamless data flow among hundreds of partners. Yet that very interdependence makes industries vulnerable to disruption. When one link fails—through a cyberattack, system error, or delayed software update—the ripple effect can paralyse global trade.

A 2023 World Bank report found that more than 60 percent of international manufacturers rely on interconnected vendor platforms with limited cybersecurity audits. Experts warn that many firms invest heavily in automation but underinvest in cyber resilience. Dr Rina Mukherjee, a cyber-risk analyst, notes, "We have optimised every second of production time but ignored the seconds it takes for ransomware to spread."

Governments, too, have struggled to keep pace. Although several countries now require critical-infrastructure operators to conduct annual security reviews, implementation remains uneven. The result, according to the Global Cyber Index 2024, is a 40 percent rise in supply-chain intrusions since 2020. Analysts describe this as the "cumulative effect of inaction"—years of neglect that compound into systemic fragility.

The larger danger, experts argue, lies not in the breaches already known but in the undiscovered points of failure embedded deep within global networks. As organisations race toward digital transformation, the challenge is to make systems not only smarter but also stronger—so that the next innovation does not become the next vulnerability.

4. What does the phrase "**cumulative effect of inaction**" suggest in the context of the passage?

- A) The government's proactive measures against cyberattacks
- B) The growing impact of long-term neglect in cybersecurity
- C) A sudden, isolated event caused by hackers
- D) The resilience built through years of practice

5. Which statement best explains why the "**just-in-time**" model increases cyber risk?

- A) It relies on multiple disconnected servers.
- B) It demands constant connectivity among partners.
- C) It eliminates human oversight completely.
- D) It stores large quantities of spare components.

6. Which option best captures the central idea of the passage?

- A) Automation ensures both speed and safety in industries.
- B) Cyberattacks are unpredictable and unavoidable.
- C) Efficiency-driven systems can become fragile without security investment.
- D) Governments should control all private supply chains.

7. Which **idiom** best describes the situation in the sentence below? "*Whenever she was asked a question about her new novel, the author would get so excited that she would speak continuously, with her words racing and tumbling over each other in a blur.*"

- A) A word in edgewise
- B) Speak volumes
- C) Mile a minute
- D) Talk shop

8. Choose the sentence that follows the correct subject-verb agreement:
- Each of the research proposals submitted by the committee members were scrutinized by the panel before the funding announcement.
 - There is a dozen eggs and three loaves of artisanal bread needed for the charity bake sale this afternoon.
 - The majority of the electorate, despite widespread media coverage, remain unconvinced about the proposed infrastructure bill's benefits.
 - Mathematics, especially the advanced principles of calculus and topology, are essential for solving the complex computational problems facing the team.
9. Choose the correct sentence transformation from active to passive voice: The engineers realized that the extensive delay in the construction of the prototype had severely jeopardized the company's bid for the lucrative government contract.
- It was realized by the engineers that the company's bid for the lucrative government contract had been severely jeopardized by the extensive delay in the construction of the prototype.
 - That the extensive delay in the construction of the prototype had severely jeopardized the company's bid for the lucrative government contract was realized by the engineers.
 - The company's bid for the lucrative government contract has been severely jeopardized by the extensive delay in the construction of the prototype, which the engineers realized.
 - The extensive delay in the construction of the prototype was being realized to have jeopardized the company's bid for the lucrative government contract by the engineers.
10. Read the sentence carefully and answer the question below: "The new policy mandates stricter data encryption; **ergo**, all department heads must attend a security workshop next week to ensure compliance." The discourse marker "**ergo**" is used to:
- Signal a preceding cause or reason for the main statement.
 - Introduce a necessary conclusion or logical inference.
 - Soften a strong claim by presenting a less certain alternative.
 - Reiterate the central idea using different languages.
11. In which of the following sentences is the word stem used in a way that reflects its most natural and idiomatic collocation?
- The manager's sudden resignation stemmed a series of rumours about internal disputes.
 - Researchers are trying to stem the rapid spread of misinformation on social media.
 - The teacher stemmed her opinion on the firm belief that discipline is outdated.
 - The heated argument stemmed across several unrelated topics during the meeting.
12. Which word best fits the context of the sentence below? The museum director rejected the donor's elaborate marble statue, declaring it a piece of _____ artistry that was gaudy and worthless, fit for a carnival, not a historical collection.
- venerable
 - tawdry
 - auspicious
 - inchoate
13. Which meaning of the word "**censure**" best fits its use in the context of the sentence below? Despite his historically accurate reporting on the political scandal, the journalist faced an official censure from the news regulatory body for failing to disclose his own family's financial ties to one of the central figures.
- To subject material to official review for the purpose of removing objectionable content.
 - To publicly express severe disapproval or formal condemnation for a specific fault.
 - To forbid the use, publication, or public availability of certain sensitive information.
 - To provide an official estimate or statistical count of a population or resource pool.
14. How many punctuation marks are required to make the following passage grammatically accurate? The brilliant scientist Dr. Elena Rios a Nobel laureate from Argentina presented her new finding it was revolutionary but the skeptical audience was unconvinced they demanded proof I thought to myself Shes the smartest person here yet no one believes her Its frustrating What if her discovery is the answer
- 13
 - 14
 - 15
 - 16

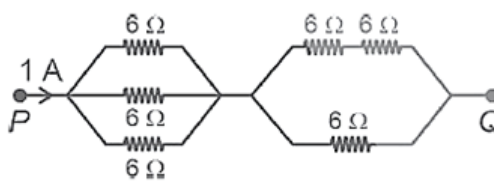
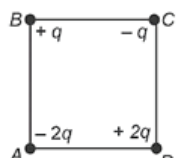
15. Identify the sentence that correctly uses a **gerund or infinitive phrase** to complete its meaning:
- The marketing team will postpone to decide on the new logo until the survey results are finalized.
 - Before sending his application, Mr. Patel had spent two hours to check every item on the long requirements list.
 - The new regulations require every employee presenting their work at the quarterly meeting.
 - Sarah chose to focus on her immediate tasks rather than worrying about future promotions.
16. Choose the option that best expresses the sentence in indirect speech: The teacher said to the class, "You must submit your assignments by Friday, or you will lose marks."
- The teacher told the class that they must submit their assignments by Friday, or they would lose marks.
 - The teacher told the class that they had to submit their assignments by Friday, or they would lose marks.
 - The teacher said to the class that you must submit your assignments by Friday, or you will lose marks.
 - The teacher said to the class that they must submit their assignments by Friday, or they will lose marks.
17. Which of the following pairs correctly shows a word and its strict antonym, emphasizing a precise contrast in attitude or action?
- Eschew – Embrace
 - Exonerate – Disown
 - Mitigate – Intensify
 - Laud – Criticize
18. Identify the clause type in the following sentence: She completed the project early because she wanted to impress her manager.
- Noun clause
 - Adjective clause
 - Adverb clause
 - Relative clause
19. Choose the option that best retains the meaning of the original sentence: Had the company foreseen the market's shift toward smaller devices, they would not have invested so heavily in large desktop technology.
- The company did not anticipate the market change, resulting in heavy investment in technology that proved inefficient.
 - Since the company invested heavily in large desktops, they were unable to foresee the coming shift in the market.
 - The company's heavy investment in large desktop technology was the direct cause of the market's eventual shift.
 - The company, which foresaw the market shift toward smaller devices, chose to invest heavily in large desktop technology anyway.
20. How many different **literary devices** are used in the following passage? The city streets were a frozen maze of fear, each shadow whispering secrets of the night."
- Two – Metaphor, Personification
 - Three – Metaphor, Simile, Alliteration
 - Two – Hyperbole, Alliteration
 - Three – Personification, Irony, Metaphor
21. Four-point charges $+q$, $-q$ and $-q$ are placed respectively at corners A, B, C and D of a square of side a . The electric potential at centre O of square is
- $\frac{1}{4\pi\epsilon_0} \frac{q}{a}$
 - $\frac{1}{4\pi\epsilon_0} \frac{2q}{a}$
 - $\frac{1}{4\pi\epsilon_0} \frac{4q}{a}$
 - Zero
22. A parallel plate capacitor have plate area A and plate separation d is charged by a battery of voltage V . The battery is then disconnected. The work needed to pull the plates to separation $2d$ is
- $\frac{AV^2\epsilon_0}{d}$
 - $\frac{2AV^2\epsilon_0}{d}$
 - $\frac{AV^2\epsilon_0}{2d}$
 - $\frac{3AV^2\epsilon_0}{2d}$
23. An electric bulb has a rating of 500W, 100V. It is used in a circuit having a 200 V supply. What resistance must be connected in series with the bulb so that it delivers 500W?
- 20 Ω
 - 50 Ω
 - 40 Ω
 - 30 Ω
24. The force between two short magnets along their line joining parallel to the axis will be inversely proportional to
- Second power of distance
 - Third power of distance
 - Fourth power of distance
 - Zero power of distance

25. An AC circuit has an inductive reactance equal to $10\ \Omega$ and a resistance of $10\ \Omega$. The effective impedance of the circuit is
 A) $20\ \Omega$ B) $10\sqrt{2}\ \Omega$ C) $10\sqrt{3}\ \Omega$ D) Zero
26. A coil of resistance $100\ \Omega$ is placed in a magnetic field. If the magnetic flux linked with the coil varies with time t as

$$\phi = 3t^2 + 2t + 4$$

 The current in the coil at $t = 3\text{ s}$ is
 A) 0.20 A B) 0.25 A C) 0.15 A D) 0.40 A
27. In the nuclear decay given below

$${}^A_Z X \rightarrow {}^A_{Z+1} Y \rightarrow {}^{A-4}_{Z-1} B \rightarrow {}^{A-4}_{Z-1} B$$

 The particles emitted in the sequence are
 A) β^-, α, γ B) γ, β^-, α C) α, β^-, γ D) β^-, γ, α
28. Light of wavelength 500 nm is used to form interference pattern in YDSE. A uniform glass plate of refractive index 1.5 and thickness 0.1 mm is introduced in path of one of interfering beams. The number of fringes which will shift will be
 A) 100 B) 120 C) 150 D) 180
29. De Broglie wavelength associated with an electron moving with speed of $5.4 \times 10^6\text{ m/s}$ is
 A) 0.35 nm B) 0.47 nm C) 0.135 nm D) 0.355 nm
30. In the displacement method, the distance between the object and screen is 70 cm and focal length of the lens is 16 cm . The separation of the magnified and diminished image positions of the lens is
 A) 17 cm B) 20.5 cm C) 25 cm D) 22.5 cm
31. Resistance of $6\ \Omega$ each are connected as shown in diagram when the current 1 A is flowing as shown. The potential difference $V_p = V_q$ is

 A) 3 V
 B) 6 V
 C) 9 V
 D) 12 V
32. If a is a distance of an object from the focus of a concave mirror of focal length f , the distance of its real image from focus is
 A) $\frac{f+a}{f}$ B) $\frac{f^2}{a}$ C) af^2 D) $\frac{af}{(a-f)}$
33. Four charges are arranged at the corners of a square ABCD as shown. The force on a $+ve$ charge kept at the centre of the square is

 A) Zero
 B) Along diagonal AC
 C) Along diagonal BD
 D) Perpendicular to the side AB
34. The coil of moving coil galvanometer is swinging periodically. To bring it to rest we have to
 A) Connect a very small resistance in parallel B) Connect a small resistance in series
 C) Earth the galvanometer case D) Bring an aluminium piece near it
35. The terminals of battery of emf 12 V and internal resistance $1\ \Omega$ are connected to a circular coil of resistance $16\ \Omega$ at two points distance a quarter of circumference of coil. The current flowing in smaller arc of circle is
 A) 2.4 A B) 2.25 A C) 0.75 A D) 1.66 A

36. Consider the following statements

- a) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ shows linkage isomerism
b) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ is an inner orbital complex
c) $\text{Cis}-[\text{CrCl}_2(\text{ox})_2]^{3-}$ is chiral and optically active species.

The correct statements is/are

- A) a) and b) only
B) b) and c) only
C) a) and c) only
D) a), b) and c)

37. Aqueous solution of which ion is colourless?

- A) Co^{2+}
B) Cu^{2+}
C) Sc^{3+}
D) V^{3+}

38. Consider the following statements

- (a) V_2O_5 is amphoteric in nature
(b) CrO and Cr_2O_3 are amphoteric oxides
(c) Mn_2O_7 is acidic in nature and green in colour

The correct statement is/are

- A) (a) only
B) (c) only
C) (a) and (b) only
D) (a) and (c) only

39. Which of the following aqueous solution will exhibit highest boiling point?

- A) 0.01 M Urea
B) 0.01 M KNO_3
C) 0.01M Na_2SO_4
D) 0.015 M $\text{C}_6\text{H}_{12}\text{O}_6$

40. Strongest acid among the following is

- A) HOCl
B) HOClO
C) HOClO_3
D) HOClO_2

41. If the half life ($t_{1/2}$) for a first order reaction is 1 minutes, then the time required for 99.9% completion of the reaction is closest to:

- A) 2 minutes
B) 4 minutes
C) 5 minutes
D) 10 minutes

42. Identify the incorrect statement from the following:

- A) The acidic strength of HX ($\text{X} = \text{F}, \text{Cl}, \text{Br}$ and I) follows the order: $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$.
B) Fluorine exhibits -1 oxidation state whereas other halogens exhibit +1, +3, +5 and +7 oxidation states also.
C) The enthalpy of dissociation of F_2 is smaller than that of Cl_2
D) Fluorine is stronger oxidising agent than chlorine.

43. Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?

- A) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
B) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
C) $[\text{Co}(\text{NH}_3)_6\text{Cl}_3]$
D) Both (a) and (b)

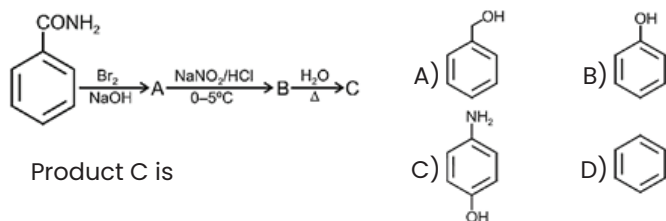
44. The compound which will react fastest by $\text{S}_\text{N}1$ mechanism is



45. Ambident nucleophile among the following is

- A) NO_2^-
B) CH_3O^-
C) H_2O
D) CH_3COO^-

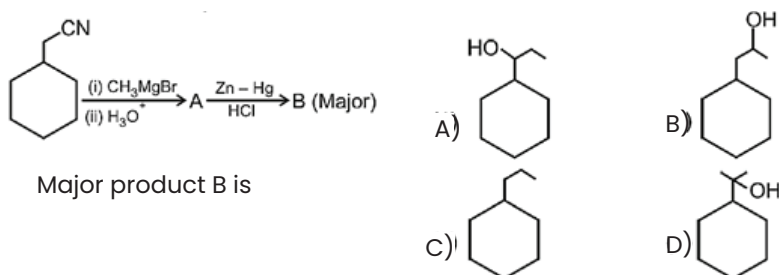
46. Consider the following reaction sequence



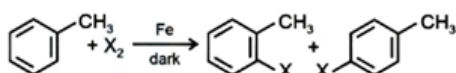
47. If rate constant of a reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$ then the time required for the completion of 90% of the reaction is

- A) 100 s B) 200 s C) 400 s D) 500 s

48. Consider the following reaction scheme



49. The following reaction method



is not suitable for the preparation of the corresponding haloarene products, due to high reactivity of halogen, when X is:

- A) F B) I C) Cl D) Br

50. Which amongst the following reaction of alkyl halides produces isonitrile as a major products?

- A) $\text{R-X} + \text{HCN} \rightarrow$ B) $\text{R-X} + \text{AgCN} \rightarrow$ C) $\text{R-X} + \text{KCN} \rightarrow$ D) $\text{R-X} + \text{NaCN}$ in $\text{C}_2\text{H}_5\text{OH}/\text{H}_2\text{O}$

Choose the most appropriate answer from the options given below

- A) (B) only B) (A) and (B) only C) (D) only D) (C) and (D) only

51. Fossil evidence shows that Homo habilis was the first to make and use tools. If researchers discover stone tools alongside a hominid skull with a cranial capacity of about 700 cc, the skull most likely belongs to:

- A) Australopithecus B) Homo habilis C) Homo erectus D) Neanderthal

52. A botanist collected a mature angiosperm seed. On cutting, he found two cotyledons and an embryo. If the endosperm was absent, the plant most likely belongs to:

- A) Maize B) Castor C) Bean D) Coconut

53. If a mutation prevents the binding of RNA polymerase to the promoter, the immediate effect will be:

- A) Inhibition of replication B) Inhibition of transcription initiation
C) Failure of RNA processing D) Termination of translation

54. In an angiosperm, mutation blocks degeneration of synergids. Which process will fail?

- A) Entry of pollen tube into ovule B) Entry of pollen tube into embryo sac
C) Release of sperm nuclei into egg cell D) Fusion of gametes inside embryo sac

55. In sewage treatment plants, activated sludge is rich in:

- A) Methanogens B) Aerobic bacteria
C) Fungi D) Cyanobacteria

56. A child bitten by a rabid dog is given both rabies vaccine and antiserum. The therapy combines:
- Active and passive immunity
 - Innate and acquired immunity
 - Natural and artificial immunity
 - Humoral and cell-mediated immunity
57. A couple prefers a natural method of family planning and chooses the "safe period" method. This is based on:
- Rhythm of menstrual cycle
 - Spermicidal action of cervical secretions
 - Hormonal imbalance in luteal phase
 - Artificial suppression of ovulation
58. A crop plant shows low yield despite abundant pollinators. Microscopic study shows non-functional tapetum. Which is the most logical reason?
- Pollen grains lack nutritive reserves
 - Pollen grains fail to reach stigma
 - Ovule is not receptive
 - Fertilisation occurs but seeds abort
59. A new fossil species is discovered with the following features:
 Upright posture and use of fire
 Cranial capacity ~950 cc
 Age: around 1.5 million years ago
 Which among the following is the closest match?
- Australopithecus
 - Homo habilis
 - Homo erectus
 - Neanderthal
60. A scientist inserts DNA into *E. coli* but finds that the inserted gene is not expressed. Later, he realizes that a promoter sequence was missing. This proves that promoters are essential for:
- Termination of transcription
 - Initiation of transcription
 - Splicing of mRNA
 - Replication of DNA
61. A tropical plain and a cold mountainous village set up identical biogas plants. The tropical plant produces more gas. The reason is:
- Higher methane content in tropical waste
 - More active methanogenic archaea in warm conditions
 - More organic waste in tropical regions
 - Absence of oxygen in plains
62. If the luteinizing hormone (LH) surge is artificially blocked in a woman, which of the following will be most directly affected?
- Follicular growth
 - Ovulation
 - Corpus luteum maintenance
 - Estrogen secretion
63. If a gene has 3000 base pairs, how many amino acids will the corresponding polypeptide have (excluding stop codon)?
- 999
 - 1000
 - 3000
 - 1500
64. Species–area curves are constructed for two regions. Region A shows a steep slope ($z = 0.3$), while Region B shows a gentle slope ($z = 0.1$). Which conclusion is correct?
- Region A is tropical, Region B is temperate
 - Region A has fewer endemic species
 - Region B has greater extinction risk
 - Both regions have similar richness
65. In sewage treatment, if secondary effluent is directly released into water bodies without further treatment, the most likely result is:
- Increase in BOD of receiving water
 - Decrease in microbial population
 - Decrease in eutrophication
 - Increase in dissolved oxygen
66. A farmer reports bollworm resistance to Bt cotton after several years of use. Which strategy prevents such resistance?
- Continuous use of same Bt gene
 - Refuge strategy with non-Bt crops
 - Application of chemical pesticides
 - Gene silencing in insect gut
67. A child shows symptoms of recurrent infections and genetic testing reveals an ADA enzyme deficiency. Which advanced therapeutic strategy can provide a permanent cure?
- Regular enzyme replacement therapy
 - Bone marrow transplantation
 - First-generation recombinant vaccine
 - Gene therapy using retroviral vector

68. In an experimental set-up, pollen grains fail to germinate on the stigma of the same species due to self-incompatibility. Which mechanism best explains this failure?
- Pollen-tube growth inhibition by stigma proteins
 - Absence of synergids in embryo sac
 - Lethal gene activation in ovary
 - Failure of karyogamy during fertilisation
69. A biotechnology company produces large amounts of human insulin using recombinant DNA. Which organism is most likely the production host?
- Bacillus subtilis*
 - Saccharomyces cerevisiae*
 - E. coli*
 - Aspergillus niger*
70. A case study showed reduced birth rates in a village where women were given vocational training along with contraceptive access. The most valid inference is:
- Economic empowerment delays childbearing
 - Contraceptive quality improved
 - Male sterilisation rates increased
 - Fertility reduction is unrelated to women's empowerment

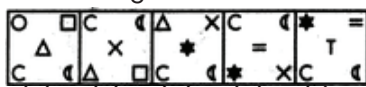
Read the following information carefully and answer the questions given below. (71 & 72)

Six boys A, B, C, D, E and F are marching in a line, their positions arranged according to their heights, the tallest at the back. F is between A and B. E is shorter than D but taller than C who is taller than A. E and F have two boys between them. A is not the shortest.

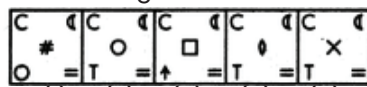
71. Where is E?
- Between A and B
 - Between A and C
 - Between C and D
 - In front of C
72. Who is shortest?
- E
 - B
 - C
 - F
73. Ashish leaves his house at 20min to seven in the morning, reaches Kunal's house in 25min, they finish their breakfast in another 15 min and leave for their office which takes another 35min. At what time do they leave Kunal's house to reach their office?
- 7:40 am
 - 7:20 am
 - 7:45 am
 - 8:15 am

74. Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

Problem Figures:



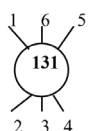
Answer Figures:



- (5)
- (4)
- (3)
- (2)

75. The positions of how many digits in the number 2451479638 will remain same when the first half and the second half of the digits are arranged in ascending order separately?
- Only one
 - Two
 - None
 - Three

76.



- 320
- 274
- 262
- 132

77. X – Z means X is the mother of Z; X x Z means X is the father of Z and X+Z means X is the daughter of Z. Now, if M – N x T+Q, then which of the following is not true?
- T is N's daughter
 - N is wife of Q
 - M is mother-in-law of Q
 - Q is wide on N

78. In a certain code language 'DLEGYLC' is written as 'ENGLAND'. How will 'HLBDY' be written in that code?
 A) INDIA B) INDFZ C) NDOIA D) JNDEA
79. In a row of boys, Karthik is seventh from the left and Sufi is twelfth from the right. If they interchange their positions, Karthik becomes twenty second from the left. How many boys are there in the row?
 A) 19 B) 29 C) 31 D) 33
80. Complete the series : 1, 9, 17, 33, 49, 73,
 A) 97 B) 98 C) 99 D) 100
81. A square paper is folded into half diagonally, and then along the midline again. If a small triangle is cut at the closed corner, what figure appears when unfolded?
 A) One triangle B) Two triangles C) Four triangles D) A star-like pattern
82. If the mirror is placed along a diagonal of a square, what happens to the square's image?
 A) No change
 B) It overlaps exactly
 C) It becomes a rectangle
 D) It flips to the other side
83. A pyramid has a square base. How many vertices does it have?
 A) 4 B) 5 C) 6 D) 8
84. **Fire : Heat :: Rain : ?**
 A) Cloud B) Thunder C) Flood D) Wetness
85. A cube is placed such that its edge is along each of the three coordinate axes (x, y, z). How many faces of the cube will be visible from the front view?
 A) 1 B) 2 C) 3 D) 4
86. What is the only known animal species considered biologically immortal (meaning it can revert to a juvenile form and restart its life cycle)?
 A) Axolotl B) Turritopsis dohrnii
 C) Greenland Shark D) Hydra
87. Which country has won the FIFA World Cup the most number of times?
 A) Argentina B) Germany C) Italy D) Brazil
88. At which session did the Indian National Congress adopt the resolution of 'Poorna Swaraj' (Complete Independence)?
 A) Bombay Session 1920 B) Lahore Session 1929
 C) Karachi Session 1931 D) Calcutta Session 1933
89. Which social media platform is primarily best suited for professional networking and job hunting?
 A) Facebook B) Instagram C) LinkedIn D) TikTok
90. In mass communication and advertising, the word "media" is often used as a singular collective noun. The original Latin singular form of "media" is:
 A) Median B) Medius C) Medium D) Intermediate
91. Which is the largest island in the world (excluding continents)?
 A) Madagascar B) Borneo C) Greenland D) New Guinea
92. In which field did Amartya Sen win the Nobel Prize?
 A) Physics B) Chemistry C) Economics D) Peace
93. Which article of the Indian Constitution guarantees the Right to Equality?
 A) Article 19 B) Article 21 C) Article 14 D) Article 32
94. At what temperature are Celsius and Fahrenheit scales equal?
 A) 0° B) 100° C) -40° D) -2730

95. What are *Glenochrysa zeylanica* and *Indophanes barbara*, recently discovered in Kerala?
A) New species of fungi B) Neuroptera species
C) Types of bacteria D) Newly identified marine mammals
96. Who is considered the “Father of the Paralympic Movement” for his work with injured World War II veterans?
A) Pierre de Coubertin B) Antonio Maglio
C) Sir Ludwig Guttman D) Juan Antonio Samaranch
97. Which term refers to the study of speech sounds of a given language and their function within that language’s sound system?
A) Phonetics B) Phonology C) Syntax D) Morphology
98. Which Indian-born British author received the Dayton Peace Prize Lifetime Achievement Award in November 2025?
A) Arundhati Roy B) V.S. Naipaul C) Jhumpa Lahiri D) Salman Rushdie
99. Which company developed the “Arattai” messaging application?
A) Reliance Jio B) TCS (Tata Consultancy Services) C) Zoho Corporation D) Infosys
100. What is considered a core value in healthy relationships?
A) Suspicion B) Control C) Trust D) Superiority